Structures and Materials

Tutorial | Dr.-Ing. Andreas Hermanutz

Introduction

1. Linear Algebra

Task 1: Given are the vector $\boldsymbol{u}$ and $\boldsymbol{v}$ and the matrix **A**

$$\vec{u} = [1.0 \quad 2.5 \quad 2.0]^T \qquad \vec{v} = [0 \quad 1.0 \quad 0.0]^T \qquad A = \begin{bmatrix} 1.0 & 1.5 & 0.0 \\ 1.5 & 2.5 & 1.0 \\ 0.0 & 1.0 & 3.0 \end{bmatrix}$$

A) Calculate the vector $\vec{c} = 2 \cdot \vec{u} - \vec{v}$
B) Calculate the magnitude of vector $\vec{u}$
C) Calculate the vector $\vec{d} = \vec{u} \times \vec{v}$
D) Calculate the vector $\vec{e} = [A]\vec{v}$, and $\vec{f} = [A]^T\vec{v}$

1. Linear Algebra

$$\vec{u} = [1.0 \quad 2.5 \quad 2.0]^T \qquad \vec{v} = [0 \quad 1.0 \quad 0.0]^T \qquad A = \begin{bmatrix} 1.0 & 1.5 & 0.0 \\ 1.5 & 2.5 & 1.0 \\ 0.0 & 1.0 & 3.0 \end{bmatrix}$$

A)

$$2 \begin{bmatrix} 1.0 \\ 2.5 \\ 2.0 \end{bmatrix} - \begin{bmatrix} 0.0 \\ 1.0 \\ 0.0 \end{bmatrix} = \begin{bmatrix} 2.0 \\ 4.0 \\ 4.0 \end{bmatrix}$$

B)

$$|\boldsymbol{u}| = \sqrt{1^2 + 2.5^2 + 2^2} = \sqrt{1 + 6.25 + 4} = \frac{3\sqrt{5}}{2} \approx 3.354$$

1. Linear Algebra

$$\vec{u} = [1.0 \quad 2.5 \quad 2.0]^T \qquad \vec{v} = [0 \quad 1.0 \quad 0.0]^T \qquad A = \begin{bmatrix} 1.0 & 1.5 & 0.0 \\ 1.5 & 2.5 & 1.0 \\ 0.0 & 1.0 & 3.0 \end{bmatrix}$$

C)

$$\vec{d} = \begin{bmatrix} \mathbf{1.0} \\ \mathbf{2.5} \\ \mathbf{2.0} \end{bmatrix} \times \begin{bmatrix} \mathbf{0.0} \\ \mathbf{1.0} \\ \mathbf{0.0} \end{bmatrix} = \begin{bmatrix} \mathbf{2.5 \cdot 0.0 - 1.0 \cdot 2.0} \\ \mathbf{2.0 \cdot 0.0 - 1.0 \cdot 0.0} \\ \mathbf{1.0 \cdot 1.0 - 2.5 \cdot 0.0} \end{bmatrix} = \begin{bmatrix} \mathbf{-2.0} \\ \mathbf{0.0} \\ \mathbf{1.0} \end{bmatrix}$$

D)

$$\vec{e} = \begin{bmatrix} 1.0 & 1.5 & 0.0 \\ 1.5 & 2.5 & 1.0 \\ 0.0 & 1.0 & 3.0 \end{bmatrix} \begin{bmatrix} \mathbf{0.0} \\ \mathbf{1.0} \\ \mathbf{0.0} \end{bmatrix} = \begin{bmatrix} \mathbf{1.0} \\ \mathbf{1.5} \\ \mathbf{0.0} \end{bmatrix} \cdot \mathbf{0.0} + \begin{bmatrix} \mathbf{1.5} \\ \mathbf{2.5} \\ \mathbf{1.0} \end{bmatrix} \cdot \mathbf{1.0} + \begin{bmatrix} \mathbf{0.0} \\ \mathbf{1.0} \\ \mathbf{3.0} \end{bmatrix} \cdot \mathbf{0.0} = \begin{bmatrix} \mathbf{1.5} \\ \mathbf{2.5} \\ \mathbf{1.0} \end{bmatrix}$$

$$A = A^T \longrightarrow \vec{f} = \vec{e}$$

1. Linear Algebra - Python

Task 2: Check your result of task 1 using Python (https://colab.research.google.com/).

```
[ ]: import numpy as np

[ ]: u = np.array([1.0, 2.5, 2.0])
     print(u)

[ ]: v = np.array([0.0, 1.0, 0.0])
     print(v)

[ ]: A = np.array([[1.0, 1.5, 0.0],[1.5,2.5,1.0],[0.0,1.0,3.0]])
     print(A)

[ ]: c=2*u-v
     print(c)

[ ]: mag_u = np.linalg.norm(u)
     print(mag_u)

[ ]: d = np.cross(u,v)
     print(d)

[ ]: e=np.matmul(A,v)
     print(e)

[ ]: f=np.matmul(A.T,v)
     print(f)
```

1. Linear Algebra

Task 3:

$$[Q] = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \qquad \vec{u} = [1 \quad 0]^T \qquad [T] = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \qquad \vec{c} = [0 \quad 2]^T$$

A) Determine the vector $\vec{v}$ resulting from the rotation of the vector $\vec{u}$ by an angle of 90° ($[Q]\vec{u}$).

$$\begin{bmatrix} \cos(90) & -\sin(90) \\ \sin(90) & \cos(90) \end{bmatrix} \begin{Bmatrix} 1 \\ 0 \end{Bmatrix}$$

$$\begin{bmatrix} \mathbf{0} & \mathbf{-1.0} \\ \mathbf{1.0} & \mathbf{0} \end{bmatrix} \begin{Bmatrix} 1 \\ 0 \end{Bmatrix} = \begin{bmatrix} \mathbf{0} \\ \mathbf{1.0} \end{bmatrix} \mathbf{1} + \begin{bmatrix} \mathbf{-1.0} \\ \mathbf{0} \end{bmatrix} \mathbf{0} = \begin{Bmatrix} \mathbf{0} \\ \mathbf{1} \end{Bmatrix}$$

B.1) Determine the vector $\vec{v}$ resulting from the rotation of the vector $\vec{u}$ by an angle of 45°.

$$\begin{bmatrix} \cos(\mathbf{45}) & -\sin(\mathbf{45}) \\ \sin(\mathbf{45}) & \cos(\mathbf{45}) \end{bmatrix} \begin{Bmatrix} 1 \\ 0 \end{Bmatrix} \qquad \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{Bmatrix} 1 \\ 0 \end{Bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix} \mathbf{1} + \begin{bmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix} \mathbf{0} = \begin{Bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{Bmatrix}$$

B.2) Has the vector's magnitude changed? Proof your conclusion.

$$\vec{v} = \sqrt{\left(\frac{\sqrt{2}}{2}\right)^2 + \left(\frac{\sqrt{2}}{2}\right)^2} = \sqrt{\frac{1}{2} + \frac{1}{2}} = \sqrt{1} = 1$$

C) Determine the vector $\vec{v}$ resulting from $[T]\vec{u}$. How can this transformation be formally described

$$\begin{bmatrix} \mathbf{2} & \mathbf{0} \\ \mathbf{0} & \mathbf{2} \end{bmatrix} \begin{Bmatrix} 1 \\ 0 \end{Bmatrix} = \begin{bmatrix} \mathbf{2} \\ \mathbf{0} \end{bmatrix} \mathbf{1} + \begin{bmatrix} \mathbf{0} \\ \mathbf{2} \end{bmatrix} \mathbf{0} = \begin{Bmatrix} \mathbf{2} \\ \mathbf{0} \end{Bmatrix}$$

vector $\vec{u}$ and $\vec{v}$ have the same direction, but vector $\vec{v}$ is twice the magnitude → stretching

D) Determine the resultant vector $\vec{v}$ after applying a sequence of transformations described by using a rotation by 45°, stretching and adding a vector $\vec{c}$: $[T][Q]\vec{u} + \vec{c}$.

$$\begin{bmatrix} \mathbf{2} & \mathbf{0} \\ \mathbf{0} & \mathbf{2} \end{bmatrix} \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{Bmatrix} 1 \\ 0 \end{Bmatrix} + \begin{Bmatrix} \mathbf{0} \\ \mathbf{2} \end{Bmatrix} = \begin{bmatrix} \mathbf{2} & \mathbf{0} \\ \mathbf{0} & \mathbf{2} \end{bmatrix} \begin{Bmatrix} \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{Bmatrix} + \begin{Bmatrix} \mathbf{0} \\ \mathbf{2} \end{Bmatrix} = \begin{bmatrix} \mathbf{2} \\ \mathbf{0} \end{bmatrix} \frac{\sqrt{2}}{2} + \begin{bmatrix} \mathbf{0} \\ \mathbf{2} \end{bmatrix} \frac{\sqrt{2}}{2} + \begin{Bmatrix} \mathbf{0} \\ \mathbf{2} \end{Bmatrix}$$

$$\begin{bmatrix} \sqrt{2} \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ \sqrt{2} \end{bmatrix} + \begin{Bmatrix} 0 \\ 2 \end{Bmatrix} = \begin{Bmatrix} \sqrt{2} \\ \sqrt{2} + 2 \end{Bmatrix}$$

1. Linear Algebra

Given is the vector $\vec{u}$ and the matrix $[A]$

$$\vec{u} = [2 \quad 3]^T \qquad [A] = \begin{bmatrix} 1.0 & 5.5 \\ 5.5 & 2.0 \end{bmatrix}$$

Task 4: Solve the linear system $[A]b = u$ and double check your solution

$$\vec{b} = [A]^{-1}\vec{u} \qquad [A]^{-1} = \frac{1}{1.0 \cdot 2.0 - 5.5^2}\begin{bmatrix} 2.0 & -5.5 \\ -5.5 & 1.0 \end{bmatrix}$$

$$\vec{b} = \frac{1}{1.0 \cdot 2.0 - 5.5^2}\begin{bmatrix} 2.0 & -5.5 \\ -5.5 & 1.0 \end{bmatrix}\begin{Bmatrix} 2 \\ 3 \end{Bmatrix} = \frac{1}{1.0 \cdot 2.0 - 5.5^2}\left(\begin{bmatrix} 2 \\ -5.5 \end{bmatrix}2 + \begin{bmatrix} -5.5 \\ 1.0 \end{bmatrix}3\right)$$

$$\vec{b} = -\frac{4}{113}\begin{bmatrix} -12.5 \\ -8 \end{bmatrix}$$

Test: $-\frac{4}{113}\begin{bmatrix} 1.0 & 5.5 \\ 5.5 & 2.0 \end{bmatrix}\begin{bmatrix} -12.5 \\ -8 \end{bmatrix} = -\frac{4}{113}\left(\begin{bmatrix} 1.0 \\ 5.5 \end{bmatrix}(-12.5) + \begin{bmatrix} 5.5 \\ 2.0 \end{bmatrix} - 8\right) = -\frac{4}{113}\begin{bmatrix} -\frac{113}{2} \\ -\frac{339}{4} \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

Task 4.1: Check the result using Python. **Option 1:**

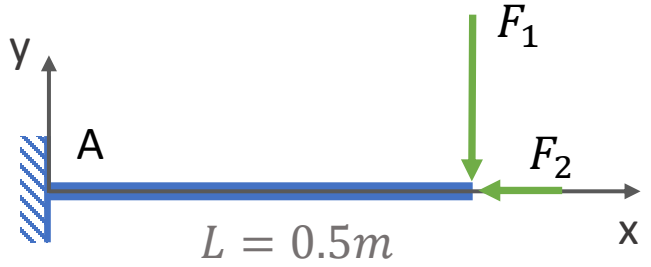
```
b=np.linalg.solve(A, u)
```

Option 2:

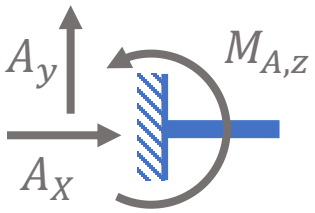
```
A_inv = np.linalg.inv(A)
```

2. Engineering Mechanics

Task 5: Given is a rigid beam loaded by a constant force $F_1 = 10N$ and $F_2 = 1N$.



A) Calculate the reaction forces.



A)

$$\sum F_x = 0: \quad A_x - F_2 = 0$$
$$A_x = F_2 = 1N$$

$$\sum F_y = 0: \quad A_y - F_1 = 0$$
$$A_y = F_1 = 10N$$

$$\sum M_z = 0: \quad M_{A,z} - F_1 \cdot L = 0$$
$$M_{A,z} = F_1 \cdot L = 5Nm$$

$$r_{AF} = \begin{bmatrix} 0.5 \\ 0.0 \end{bmatrix} \qquad F = \begin{bmatrix} -1.0 \\ -10.0 \end{bmatrix}$$

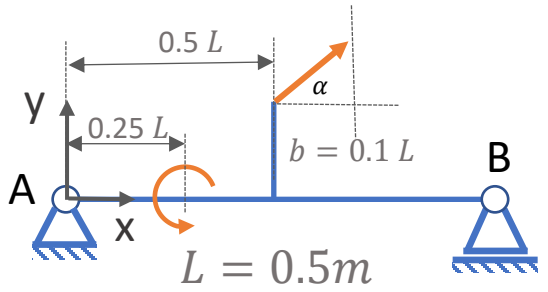
$$F_A + F = 0 \qquad \begin{bmatrix} A_x \\ A_y \end{bmatrix} + \begin{bmatrix} -1.0 \\ -10.0 \end{bmatrix} = 0$$

$$\rightarrow \begin{bmatrix} A_x \\ A_y \end{bmatrix} = \begin{bmatrix} 1.0 \\ 10.0 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 0 \\ M_{A,z} \end{bmatrix} + \begin{bmatrix} 0.0 \\ 0.5 \\ 0.0 \end{bmatrix} \times \begin{bmatrix} -1.0 \\ -10.0 \\ 0.0 \end{bmatrix} = 0 \rightarrow \begin{bmatrix} 0 \\ 0 \\ M_{A,z} \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.0 \\ 0.0 \end{bmatrix} \times \begin{bmatrix} 1.0 \\ 10.0 \\ 0.0 \end{bmatrix} = \begin{bmatrix} 0.0 \\ 0.0 \\ 5.0 \end{bmatrix}$$

2. Engineering Mechanics

Task 6: Given is a rigid frame structure loaded by a constant force $\mathrm{F} = 10N$ and a moment $M_1 = 12Nm, \alpha = 45deg$



A) Calculate the reaction forces.

$$F_1 = 10\begin{bmatrix}\cos 45\\ \sin 45\end{bmatrix} = 10\begin{bmatrix}\frac{\sqrt{2}}{2}\\ \frac{\sqrt{2}}{2}\end{bmatrix} = 5\begin{bmatrix}\sqrt{2}\\ \sqrt{2}\end{bmatrix}$$

$$r_{AF} = L\begin{bmatrix}0.5\\ 0.1\end{bmatrix}$$

$$r_{AB} = L\begin{bmatrix}1.0\\ 0.0\end{bmatrix}$$

$$\sum F = \begin{bmatrix}A_x\\ A_y\end{bmatrix} + \begin{bmatrix}0\\ B_y\end{bmatrix} + 5\begin{bmatrix}\sqrt{2}\\ \sqrt{2}\end{bmatrix} = 0$$

$$\sum M^{(A)} = M_1 + L\begin{bmatrix}0.5\\ 0.1\end{bmatrix} \times \begin{bmatrix}F_x\\ F_y\end{bmatrix} + L\begin{bmatrix}1.0\\ 0.0\end{bmatrix} \times \begin{bmatrix}0\\ B_y\end{bmatrix} = 0$$

$$0 = M_1 + L \cdot 0.5 \cdot F_y - L \cdot 0.1 \cdot F_x + B_y L$$

$$A_x = -5\sqrt{2}$$

$$A_y + B_y = -5\sqrt{2}$$

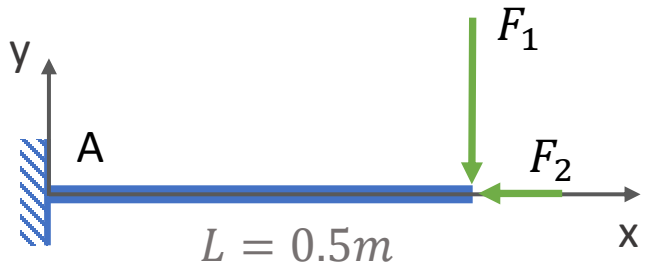
$$B_y L = -M_1 - 2L\sqrt{2}$$

$$\begin{bmatrix}1 & 1\\ 0 & L\end{bmatrix}\begin{bmatrix}A_y\\ B_y\end{bmatrix} = \begin{bmatrix}-5\sqrt{2}\\ -M_1 - 2L\sqrt{2}\end{bmatrix}$$

$$\begin{bmatrix}A_y\\ B_y\end{bmatrix} = \frac{1}{L}\begin{bmatrix}L & -1\\ 0 & 1\end{bmatrix}\begin{bmatrix}-5\sqrt{2}\\ -M_1 - 2L\sqrt{2}\end{bmatrix} = \frac{1}{L}\begin{bmatrix}-5L\sqrt{2} + M_1 + 2L\sqrt{2}\\ -M_1 - 2L\sqrt{2}\end{bmatrix}$$

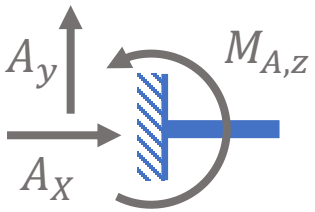
$$\begin{bmatrix}A_y\\ B_y\end{bmatrix} = \begin{bmatrix}-3\sqrt{2} + \frac{M_1}{L}\\ -\frac{M_1}{L} - 2\sqrt{2}\end{bmatrix} \approx \begin{bmatrix}19.75\\ -26.83\end{bmatrix} N$$

2. Engineering Mechanics



Task 7: Given is a rigid beam loaded by a constant force $F_1 = 10N$ and $F_2 = 1N$.

A) Calculate the reaction forces.
B) Create a sketch of the internal vertical force distribution.
C) Create a sketch of the internal moment distribution.



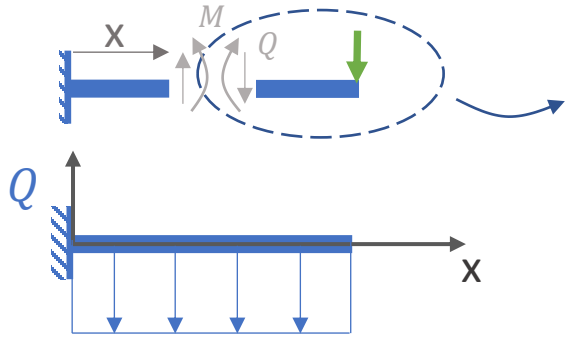
A)

$$\sum F_x = 0: \quad A_x - F_2 = 0$$
$$A_x = F_2 = 1N$$

$$\sum F_y = 0: \quad A_y - F_1 = 0$$
$$A_y = F_1 = 10N$$

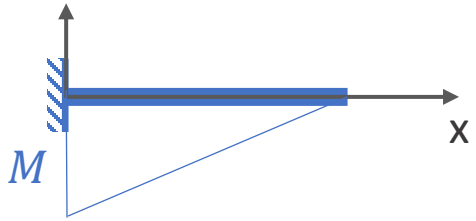
$$\sum M_z = 0: \quad M_{A,z} - F_1 \cdot L = 0$$
$$M_{A,z} = F_1 \cdot L = 5Nm$$

B)



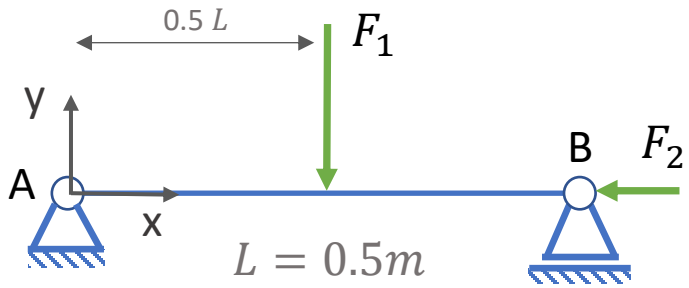
$$-Q - F_1 = 0$$
$$Q = -F_1 = -10N$$

C)



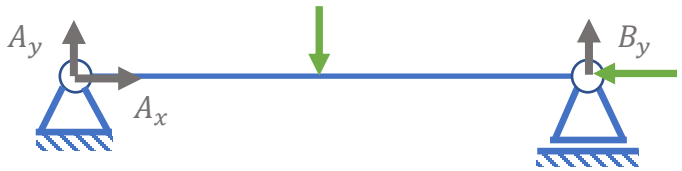
$$-M - F_1(L - x) = 0$$
$$M = F_1(x - L)$$

2. Engineering Mechanics



Task 8: Given is a rigid beam loaded by a constant force $F_1 = 10N$ and $F_2 = 1N$.

A) Calculate the reaction forces.
B) Create a sketch of the internal vertical force distribution.
C) Create a sketch of the internal moment distribution.



A)

$$\sum F_x = 0: \quad A_x - F_2 = 0$$

$$A_x = F_2 = 1N$$

$$\sum F_y = 0: \quad A_y + B_y - F_1 = 0$$

$$A_y = F_1 - B_y$$

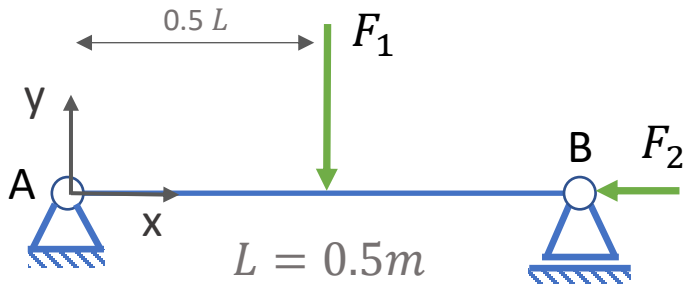
$$\sum M_z = 0: \quad -F_1 \cdot 0.5L + L \cdot B_y = 0$$

$$L \cdot B_y = F_1 \cdot 0.5L$$

$$B_y = F_1 \cdot 0.5 = 5N$$

$$A_y = F_1 - B_y = 5N$$

2. Engineering Mechanics

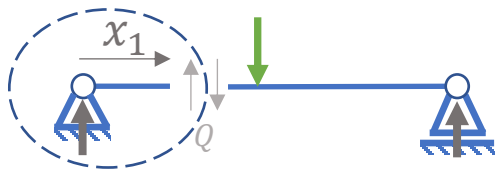


Task 8: Given is a rigid beam loaded by a constant force $F_1 = 10N$ and $F_2 = 1N$.

A) Calculate the reaction forces.
B) Create a sketch of the internal vertical force distribution.
C) Create a sketch of the internal moment distribution.

B)

Section 1

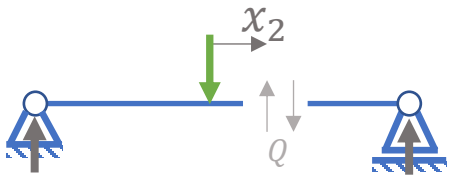


$0 < x_1 < 0.5L$

$Q + A_y = 0$

$Q = -A_y = -5N$

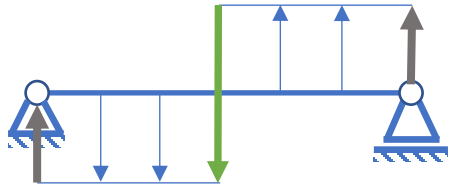
Section 2



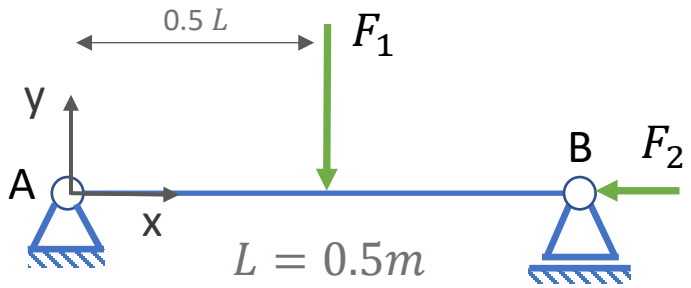
$0.5L < x_2 < L$

$Q + A_y - F_1 = 0$

$Q = -A_y + F_1 = 5N$



2. Engineering Mechanics

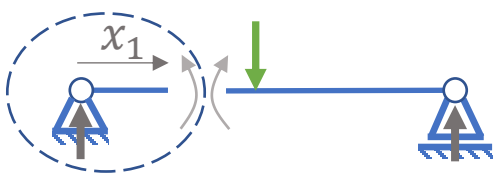


Task 8: Given is a rigid beam loaded by a constant force $F_1 = 10N$ and $F_2 = 1N$.

A) Calculate the reaction forces.
B) Create a sketch of the internal vertical force distribution.
C) Create a sketch of the internal moment distribution.

C)

Section 1

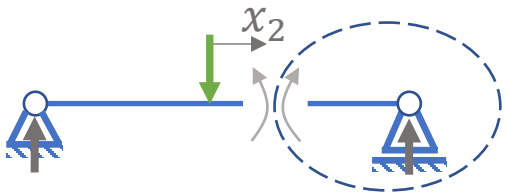


$0 < x < 0.5L$

$M - A_y x_1 = 0$

$M = A_y x_1 = 5x_1$

Section 2



$0.5L < x < L$

$-M + B_y\left(\frac{L}{2} - x_2\right) = 0$

$M = B_y\left(\frac{L}{2} - x_2\right) = 0$

$M(x_1 = 0.5L) = 5\frac{L}{2}Nm$

$M(x_2 = 0) = 5\frac{L}{2}Nm$

